Lynx at the NADI 2026 Shared Task: Exploring ASR Representations and Transcripts for Dialectal Arabic Speech Understanding

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Abstract

Speech conveys complementary information through its linguistic content and acoustic characteristics. Speech-related tasks can exploit either source of information or combine them to improve performance. In this paper, we present ***Lynx_Id*** and ***Lynx_In***, our systems for the ***NADI 2026*** Spoken Dialect Identification and Intent Recognition tasks, respectively. Both systems build on an automatic speech recognition (ASR) model. *Lynx_Id* is a late-fusion system for dialect identification. It combines internal representations from the ASR encoder with predictions derived from the ASR transcript. *Lynx_In* is a two-branch text-based system that uses ASR transcriptions for intent recognition. One branch predicts the intent among the known classes, while the second acts as a gate to detect out-of-scope utterances. *Lynx_Id* achieved 53.53% accuracy and ranked third in the Spoken Dialect Identification task, while *Lynx_In* achieved a weighted F1 score of 76.86% and ranked first in the Intent Recognition task.

1 Introduction

Speech-based Arabic language processing is particularly challenging because dialectal variation affects both pronunciation and lexical choice. In addition, spoken-language systems must cope with differences in recording conditions, domains, and ASR errors. These challenges make it useful to consider both the acoustic information encoded by speech models and the lexical information available through their transcriptions.

Beyond transcription, recent work has explored the internal encoder representations of ASR models for downstream tasks (Gong et al., 2023; Radhakrishnan et al., 2023; Lugosch et al., 2019). These representations capture information from the speech signal, while the decoded transcripts provide a textual view that can be processed using standard NLP models (Pasad et al., 2021; Arora et al., 2024). Depending on the downstream task, these two sources of information can be exploited independently or combined (Yoon et al., 2018; Ali et al., 2016; Serdyuk et al., 2018).

Arabic dialects exhibit substantial variation in both their acoustic characteristics and lexical content, which makes dialectal speech understanding challenging, particularly under domain shift and open-set conditions. NADI 2026 (Sullivan et al., 2026) addresses these challenges through a set of tasks on robust and multidialectal Arabic speech processing. In this work, we participate in Task 2, Spoken Dialect Identification, which requires identifying the Arabic variety of a speech segment under cross-domain conditions, and Task 5.1, Intent Recognition, which requires predicting the intent of Tunisian Arabic speech while detecting utterances outside the known scenarios.

We present *Lynx_Id* and *Lynx_In* for these two tasks, respectively. Both systems use the same frozen ASR backbone but exploit it differently. *Lynx_Id* combines acoustic encoder representations with lexical predictions derived from ASR transcripts, while *Lynx_In* relies on the transcripts and separates intent classification from out-of-scope detection. *Lynx_Id* ranked third in Spoken Dialect Identification, while *Lynx_In* ranked first in Intent Recognition.

2 Data

We used different datasets for the two NADI 2026 tasks. For Spoken Dialect Identification, we used ADI-20 (Elleuch et al., 2025) and the Casablanca dataset (Talafha et al., 2024). For Intent Recognition, we used SLURP-TN (Elleuch et al., 2026) and additional data from MASSIVE (FitzGerald et al., 2023). The following subsections describe the data used for each task.

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2.1 Spoken Dialect Identification

For Spoken Dialect Identification, we used ADI-20 (Elleuch et al., 2025), an extension of the ADI-17 dataset (Ali et al., 2019). ADI-20 covers 19 Arabic dialects in addition to Modern Standard Arabic (MSA). The full corpus contains 3,556 hours of speech distributed across the different dialects.

As the main test set differs in domain and source from the training and development data, we used a sample from the Casablanca dataset as additional out-of-domain data to favour configurations that generalize across domains and to assess the cross-domain generalization of our models. The sample contains 2,000 speech segments from eight Arabic dialects: Algerian, Egyptian, Jordanian, Mauritanian, Moroccan, Palestinian, Emirati, and Yemeni. We used one subset as a secondary-domain validation set to guide hyperparameter selection for the fusion and adaptation components, and a separate random subset of 739 samples as a test set for the ablation study.

2.2 Intent Recognition

For Intent Recognition, we used SLURP-TN as the main dataset. SLURP-TN contains 2,677 training utterances and 595 validation utterances in Tunisian Arabic. The dataset covers six scenarios: alarm, email, general, news, takeaway, and weather, with 14 intent classes. Since the shared-task test set contains samples from scenarios that are not present in SLURP-TN, the system must predict one of the known intents when applicable or assign the label `unknown` otherwise.

To provide examples from additional scenarios, we also used the Arabic portion of MASSIVE. MASSIVE covers 18 scenarios, including the six scenarios found in SLURP-TN and 12 additional scenarios. We used 11,514 utterances from its training split. Since these utterances are in Modern Standard Arabic, we translated them into Tunisian Arabic using *gemma-3-27b-it* (Team et al., 2025). The generated data were used to train the out-of-scope detection component and to augment the intent classifier with additional Tunisian Arabic examples from the known scenarios, with 15% held out for threshold calibration. For the ablation study, we used a sample of the translated MASSIVE validation split, which we refer to as TN-EVAL. TN-EVAL contains 595 examples from the six known scenarios and 1,439 examples from the 12 unseen scenarios.

3 System

3.1 ASR Backbone

Both of our systems build on *Cohere Transcribe Arabic* (`cohere-transcribe-arabic-07-2026`) (Cassini et al., 2026). The model is an Arabic-focused version of *Cohere Transcribe* (Mack et al., 2026), a 2B-parameter encoder-decoder ASR model. Its architecture consists of a Conformer encoder (Gulati et al., 2020) for acoustic representation learning and a lightweight autoregressive Transformer decoder for transcription. The model takes a speech waveform as input, converts it into a log-Mel spectrogram, and produces the corresponding text transcription.

Cohere Transcribe Arabic adapts this architecture to the characteristics of Arabic speech. Its training data cover different Arabic dialects, Arabic-English code-switching, Arabic-accented English, and speech recorded under different acoustic conditions (Cassini et al., 2026). These properties make the model particularly suitable for the NADI tasks, which involve dialectal Arabic speech and, in the Intent Recognition task, code-switched Tunisian Arabic. We keep the ASR model frozen in all experiments and exploit it in two different ways. *Lynx_Id* uses both its encoder representations and generated transcripts for Spoken Dialect Identification, while *Lynx_In* relies on the generated transcripts for Intent Recognition.

3.2 Lynx_Id: Spoken Dialect Identification

Dialectal variation can be reflected both in the acoustic properties of speech and in the lexical content of its transcription. Accordingly, *Lynx_Id* combines acoustic and lexical information obtained from the ASR backbone. The acoustic branch uses the internal encoder representations, while the lexical branch uses the decoded transcript. Their predictions are combined through late fusion. An overview of the *Lynx_Id* architecture is shown in Figure 1.

3.2.1 Acoustic Branch

For each input, the frozen encoder produces 49 layer-wise representations, each with a dimensionality of 1,280. We mean-pool each representation over the temporal dimension, resulting in the sequence

$$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{49}], \quad (1)$$

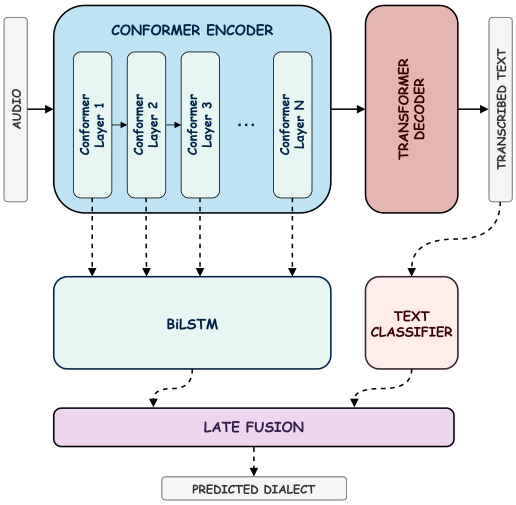


Figure 1: Overview of the *Lynx_Id* architecture for spoken dialect identification.

where $\mathbf{h}_l$ denotes the pooled representation of encoder layer l.

We treat encoder depth as a sequence and model the pooled layer representations with a single-layer bidirectional LSTM. Each 1,280-dimensional representation is normalized and projected to 256 dimensions before being passed to the BiLSTM. The final forward and backward states are concatenated and passed to a linear classification layer.

3.2.2 Lexical Branch

The lexical branch operates on the ASR transcripts. We apply light Arabic text normalization and represent the transcripts using word-level TF-IDF features over unigrams and bigrams. The resulting representations are classified using a linear support vector machine (SVM). Since the SVM outputs decision scores rather than calibrated probabilities, we apply temperature scaling before fusion.

3.2.3 Late Fusion and Prior Adaptation

We combine the acoustic and lexical posterior distributions using a weighted geometric mean:

$$P(y \mid x) \propto P_{\text{ac}}(y \mid x)^{\alpha} P_{\text{lex}}(y \mid x)^{1-\alpha}, \quad (2)$$

where P_{ac} and P_{lex} are the acoustic and lexical posterior distributions, respectively, and α is a hyperparameter that controls the contribution of the acoustic branch.

The cross-domain setting may introduce a shift in the dialect prior between the training and target data. We address this using unsupervised prior adaptation (Saerens et al., 2002) on the fused posterior distribution. The method estimates the target class prior from unlabeled predictions and adjusts the posterior probabilities accordingly. Before prior adaptation, we apply temperature scaling to the fused posterior, with temperature T controlling its sharpness.

3.3 Lynx_In: Intent Recognition

Lynx_In is a text-based model that uses the transcript generated by the ASR backbone and follows a two-branch architecture. The first branch predicts the intent among the known classes, while the second determines whether the input belongs to one of the known scenarios. The final prediction is obtained by combining the outputs of these two branches.

3.3.1 Intent Classification

The intent branch is a closed-set classifier over the known intent classes. We represent each ASR transcript using a combination of character- and word-level TF-IDF features. Character n-grams help account for orthographic variation in the ASR output, while word n-grams capture lexical cues associated with different intents. The combined representation is classified using one-vs-rest logistic regression.

3.3.2 Scenario Gate

To identify inputs that fall outside the known intent space, we use a scenario-level gate based on MARBERTv2 (Abdul-Mageed et al., 2021). We fine-tune MARBERTv2 to classify the transcript into the 18 scenarios defined in MASSIVE. Six of these scenarios correspond to those observed in SLURP-TN, while the remaining 12 represent unseen scenarios.

For an input x, the gate produces a probability distribution over the 18 scenarios. We compute the probability of belonging to the known set by summing the probabilities assigned to the six known scenarios:

$$P_{\text{known}}(x) = \sum_{s \in \mathcal{S}_{\text{known}}} P(s \mid x), \quad (3)$$

where $\mathcal{S}_{\text{known}}$ denotes the set of scenarios present in SLURP-TN. A threshold τ is then used to determine the final output:

$$\hat{y} = \begin{cases} \hat{y}_{\text{intent}}, & P_{\text{known}}(x) \geq \tau, \\ \text{unknown}, & P_{\text{known}}(x) < \tau. \end{cases} \quad (4)$$

Rank	Team	Accuracy (%)
1	Thakaa	56.15
2	Salesteq	54.66
3	*Lynx_Id* (ours)	53.53

Table 1: Official leaderboard for the Spoken Dialect Identification task.

Rank	Team	WF1 (%)
1	*Lynx_In* (ours)	76.86
2	CIS-OpenSLU	71.54
3	A3da2 Messi	69.92

Table 2: Official leaderboard for the Intent Recognition task.

where $\hat{y}_{\text{intent}}$ is the prediction of the intent classifier and τ is a hyperparameter controlling the known–unknown decision.

4 Experiments and Results

4.1 Experimental Setup

Lynx_Id. For the acoustic branch, we use AdamW with a learning rate of 10^{-3} and a batch size of 256. The model is trained for 30 epochs, and the checkpoint with the highest validation accuracy is retained. For the lexical branch, we use word unigrams and bigrams with a linear SVM with $C = 0.35$ and balanced class weights.

Lynx_In. The intent classifier combines character n-grams of length 2–5 with word unigrams and bigrams and uses one-vs-rest logistic regression with $C = 2.0$ and balanced class weights. The MARBERTv2 scenario gate is fine-tuned for four epochs using AdamW with a learning rate of 2×10^{-5} and a batch size of 32. Additional implementation details are provided in Appendix A.

4.2 Official Leaderboard Results

Tables 1 and 2 present the official leaderboard results on the shared-task test sets. For Spoken Dialect Identification, *Lynx_Id* achieved an accuracy of 53.53% and ranked third. For Intent Recognition, *Lynx_In* achieved a weighted F1 score of 76.86%, ranked first, and outperformed the second-ranked system by 5.32 percentage points.

5 Hyperparameter Selection

For *Lynx_Id*, the SVM calibration temperature is optimized by minimizing the negative log-likelihood on the ADI-20 validation set and is set to 0.310. The remaining fusion and prior-adaptation hyperparameters are selected using the secondary-domain Casablanca development set. We evaluate 201 uniformly spaced values for the fusion weight α and select $\alpha = 0.24$. Before prior adaptation, we apply temperature scaling to the fused posterior to reduce overconfidence. We select a posterior temperature of $T = 1.25$ and a minimum class-prior value of 0.002. For Cost Average, an average detection cost based on miss and false-alarm rates (Lee et al., 2022), we rescale the final logits and select a scaling factor of 1.756 on the Casablanca development set.

For *Lynx_In*, we select the known–unknown threshold τ by maximizing weighted F1 on the held-out calibration subset of the rewritten MASSIVE training data, which contains 1,727 samples. Using MARBERTv2 as the scenario gate, the selected threshold is $\tau = 0.8232$.

Configuration	Prior adapt.	Accuracy (%)
Acoustic	✗	46.28
Acoustic	✓	49.66
Lexical	✗	41.95
Lexical	✓	32.21
Acoustic + Lexical	✗	54.94
Acoustic + Lexical	✓	**62.25**

Table 3: Ablation of the LYNX_ID branches and prior adaptation.

Config.	wF1	Known R.	Unseen R.
A_1	83.89	58.15	94.16
A_2	**89.58**	**77.98**	94.16
A_3	89.39	73.78	**96.45**

Table 4: Ablation results for LYNX_IN on TN-EVAL. Known R. is recall on examples from the six known scenarios. Unseen R. is recall on examples from the out-of-scope scenarios.

5.1 Ablation Study

5.1.1 *Lynx_Id*

We ablate the acoustic branch, lexical branch, their fusion, and prior adaptation on the held-out test subset. Table 3 shows that fusion improves accuracy from 46.28% for the acoustic branch alone to 54.94%, while prior adaptation further increases it to 62.25%, confirming the complementarity of the two branches.

5.1.2 *Lynx_In*

For *Lynx_In*, we conduct a component analysis using three configurations on TN-EVAL. Configuration A_1 uses the rewritten data for the scenario

gate, A_2 additionally augments the intent classifier with known-scenario examples, and A_3 further applies the calibrated rejection threshold. We report weighted F1 and recall on known and unseen examples. Table 4 shows that A_1 to A_2 improves weighted F1 from 83.89% to 89.58% and known recall from 58.15% to 77.98%. Configuration A_3 further improves unseen recall to 96.45%, with a small decrease in weighted F1.

6 Conclusion

We presented *Lynx_Id* and *Lynx_In*, our systems for the NADI 2026 Spoken Dialect Identification and Intent Recognition tasks. Both systems build on the same frozen ASR model but exploit it differently. *Lynx_Id* combines acoustic representations from the ASR encoder with lexical information from its transcripts, and our ablation shows that the two branches provide complementary information. *Lynx_In* relies on ASR transcripts and separates intent classification from out-of-scope detection. The results show that synthetic Tunisian data substantially improve intent recognition, while threshold calibration improves the detection of unseen scenarios. *Lynx_Id* ranked third in Spoken Dialect Identification, while *Lynx_In* ranked first in Intent Recognition.

Limitations

Dependence on the ASR backbone. *Lynx_Id* and *Lynx_In* rely on the same frozen ASR model. As a result, errors in the ASR representations or generated transcripts can propagate to the downstream classifiers. This is especially relevant for *Lynx_In*, which relies entirely on the ASR transcript. In addition, we cannot determine whether its training data contain data that overlap with, or are closely related to, the datasets used in this shared task.

Quality of generated transcripts and translations. The text-based parts of our systems depend on automatically generated text. ASR transcripts may contain recognition errors, especially for dialectal and code-switched speech, and these errors can affect the downstream predictions. Similarly, the Tunisian Arabic data translated with Gemma may contain translation or rewriting errors and may not always reflect natural Tunisian usage.

Use of out-of-domain development data. We use out-of-domain data to study generalization and to select some hyperparameters. While this provides a useful proxy for domain shift, we cannot guarantee that these data reflect the distribution of the competition test set. The selected settings may therefore not transfer equally well to other domains or recording conditions.

Computational cost. The ASR backbone contains approximately 2B parameters. Although it is kept frozen, running it is still computationally expensive compared with the lightweight downstream models used in both systems. This may limit the practicality of the approach in settings with limited computational resources.

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A Additional Implementation Details

For the *Lynx_Id* acoustic branch, we use GELU activations and a dropout rate of 0.5. AdamW uses a weight decay of 10^{-4}, together with a cosine learning-rate schedule. For the lexical branch, the TF-IDF vectorizer uses a minimum document frequency of 2, a maximum document frequency of 0.9995, and a maximum vocabulary size of 400,000 features. We apply sublinear term-frequency scaling and L_2 normalization. For the *Lynx_In* intent classifier, the character and word TF-IDF representations are limited to 100,000 and 50,000 features, respectively. Both use a minimum document frequency of 2 and sublinear term-frequency scaling. Logistic regression uses the `liblinear` solver with a maximum of 1,000 iterations. For the scenario gate, we use a maximum sequence length of 64, a weight decay of 0.01, and a cosine learning-rate schedule. For synthetic data generation, we use 12 ASR transcripts from SLURP-TN as in-context examples for *gemma-3-27b-it*. Generation uses a temperature of 0.2, top-p of 0.95, a maximum of

64 generated tokens, and a batch size of 32. We used a fixed seed of 0 for all experiments.